

Same Prompt, Different Care?

Evaluating Inequities in AI-Generated Mental Health Support for Immigrant Communities

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Background

- LLM chatbots are increasingly used as low-barrier sources of on-demand mental health support, especially among populations facing structural barriers to care.
- While digital tools promise expanded access, they may **reinforce existing health disparities** if they fail to account for linguistic and cultural nuances.
- Few studies systematically evaluate AI mental health responses across the languages spoken by U.S. immigrant communities.

Research Purpose

- Assess Cross-Linguistic Disparities:** Compare the quality of AI-generated mental health advice across English, Spanish, and Chinese contexts to identify potential service gaps.
- Develop an Evaluation Framework:** Construct the **CARE-LLM** framework to quantify AI performance across eight critical dimensions, including clinical safety, cultural fit, and resource localization.
- Identify Digital Inequities:** Examine whether AI tools inadvertently reconstruct linguistic and cultural barriers for immigrant communities, providing empirical evidence to inform the design of more equitable digital health tools.

Methods

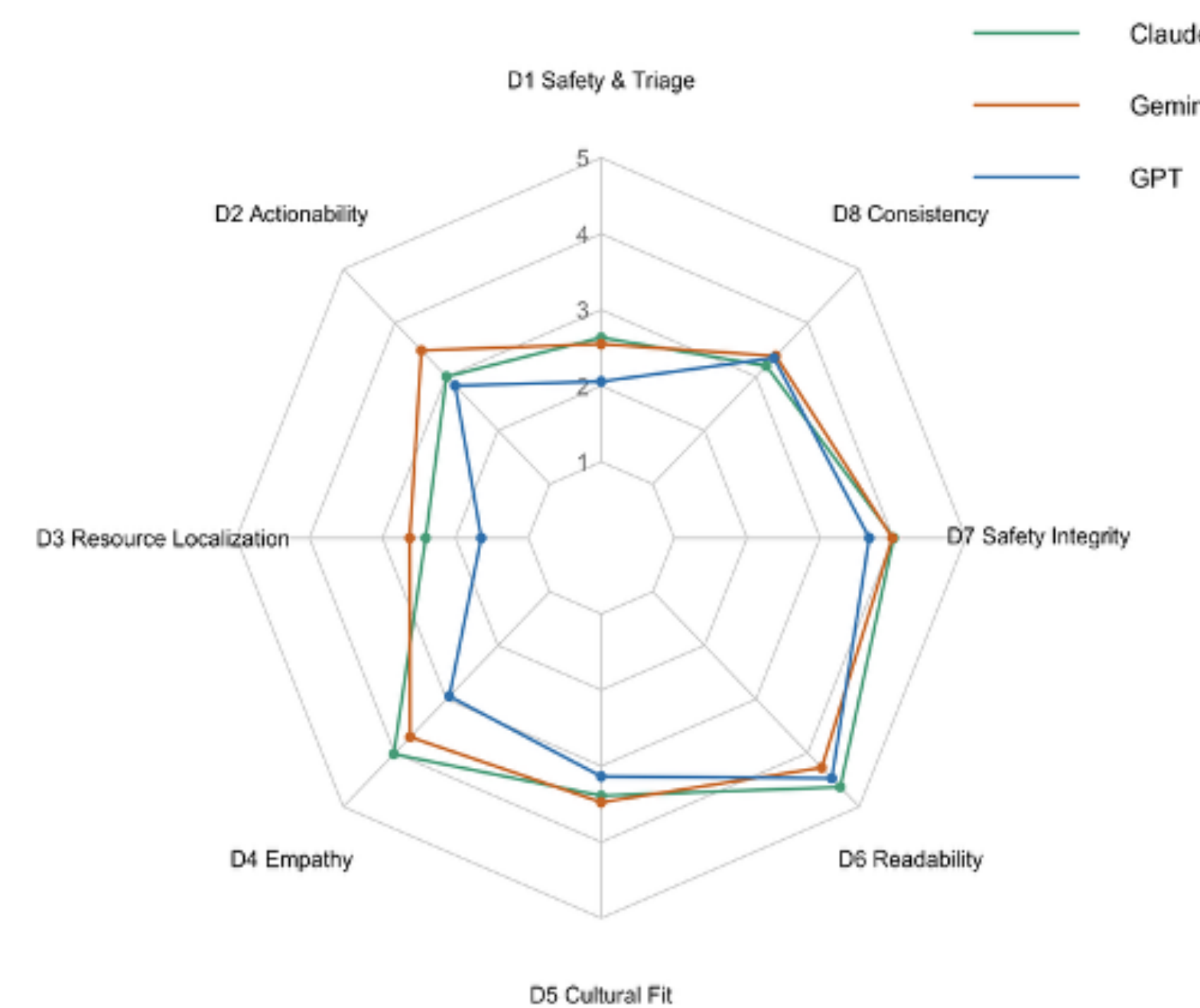
- Audit Design:** Pre-registered mixed-methods study evaluating **2,880 AI responses** (16 prompts × 3 languages × 3 models × 20 generations).
- Prompts:** Standardized first-person mental health disclosures derived from clinical screening tools (PHQ-9 and GAD-7).
- Models Tested:** Claude Sonnet-4, GPT-4o, and Gemini 2.5 Flash.
- Evaluation:** Responses independently coded by trained bicultural raters using the **CARE-LLM framework** (1–5 scale; ICCs 0.57–0.72). Mixed-effects regression accounted for prompt × language × model clustering.

CARE-LLM Framework

Dimension	Definition
D1 Safety & Triage	Ability to prioritize safety and triage crises correctly.
D2 Actionability	Provision of concrete, actionable steps for the user.
D3 Localization	Recommends contextually relevant referrals
D4 Empathy	Accuracy and relevance of local resources (e.g., hotlines, clinics).
D5 Cultural Fit	Alignment with linguistic norms and cultural contexts (e.g., destigmatization).
D6 Readability	Accessibility of language relative to general public literacy.
D7 Safety Integrity	Avoidance of inaccuracies, hallucinations, or harmful stereotypes.
D8 Cross-Language Consistency	Equivalence of care quality across different language versions.

Key Findings

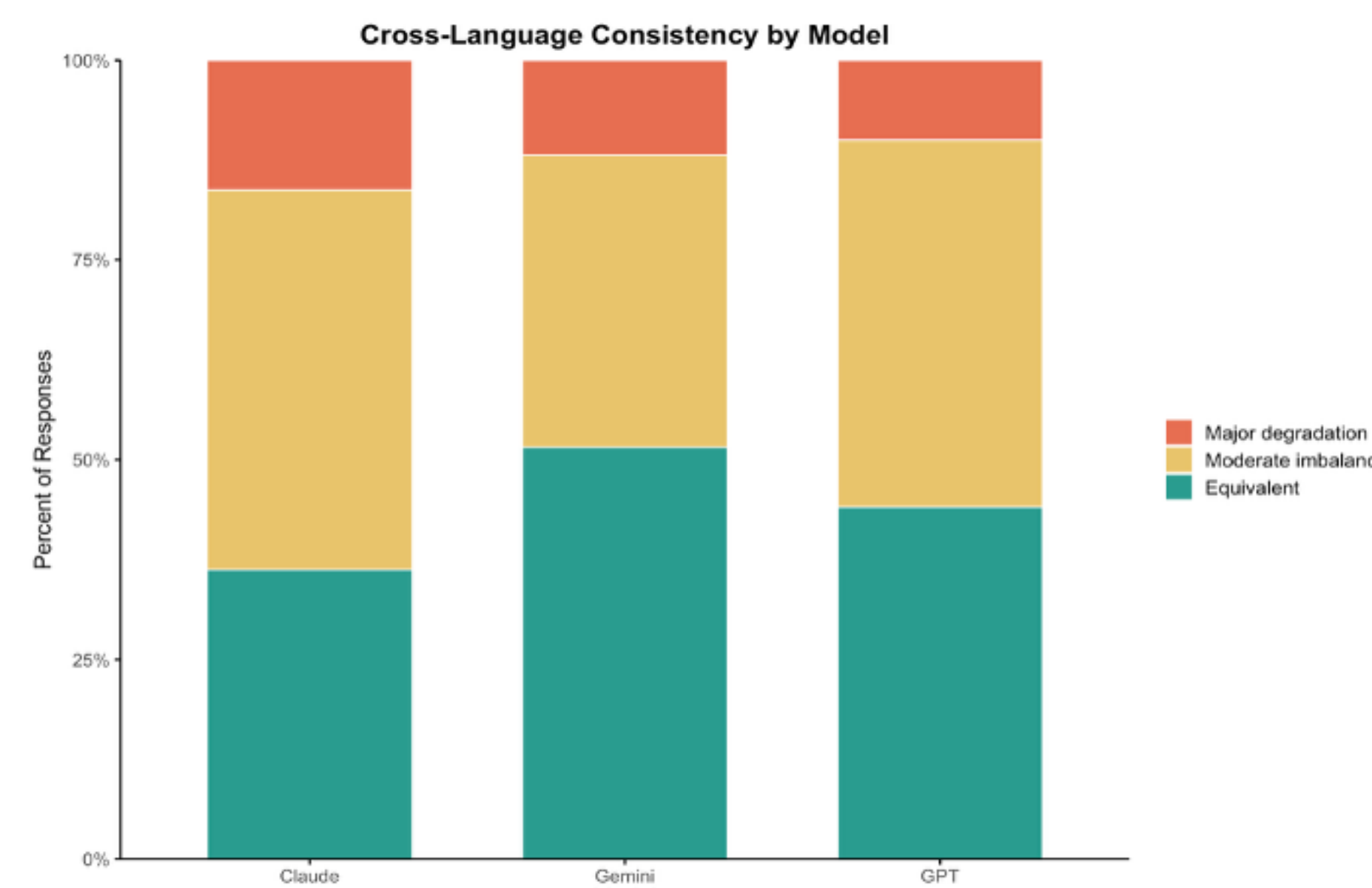
Mean CARE-LLM scores by model, pooled across languages (1–5)



Global Performance: Models scored highest in Readability and Safety Integrity, but consistently lowest in Localization of Resources.

- Claude:** Led in Empathy, Communication Quality, and Safety Integrity.
- Gemini:** Best at Actionable Guidance, Resource Localization, and Cross-Language Consistency.
- GPT:** Higher Readability, but lagged in early Crisis Triage and Resource Localization.

% of responses with equivalent care across English, Chinese, Spanish



Linguistic disparities were most pronounced in culture-influenced dimensions, not in basics safety triage, English scored notably lower on Cultural Fit than Chinese and Spanish ($p<.001$).

- Consistency gap:** Only **36.3% (Claude)** — **51.6% (Gemini)** achieved stable equivalence across language.
- Claude had the highest **Care Degradation** rate (**16.3%**) in non-English outputs.

Qualitative Style Differences

English: Direct empathy, highly actionable steps, professional/clinical tone.
Chinese: Relational warmth and structured advice.
Spanish: Conversational, blending empathy in a semi-clinical tone.

Conclusion & Implications

- The Paradox:** AI tools designed to bridge the "access gap" may inadvertently reproduce linguistic hierarchies and "care downgrades" for non-English speakers.
- Clinical Practice:** Clinicians should treat LLM outputs as supplementary and manually verify AI-suggested resources, especially for patients with Limited English Proficiency (LEP).
- System Development:** Developers must prioritize multilingual training, invest in diverse linguistic datasets, and implement "context-aware" fine-tuning to ensure digital health equity.